Melek Moalla

SOFTWARE & AI ENGINEERING STUDENT

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PROFESSIONAL PROFILE

Fourth-year Artificial Intelligence Engineering student at ESPRIT (expected graduation 2028), ranked 1st in annual class standings, with three software engineering internships and hands-on experience in real-time systems, reinforcement learning, computer vision, and agentic AI. Developer of independent AI projects and founder of an emerging software development initiative delivering interactive digital experiences and building customer loyalty solutions for local businesses.

Seeking a Summer 2027 internship in AI Engineering, Machine Learning, Software Engineering, or Agentic AI, with a particular interest in challenging, production-oriented engineering environments.

EXPERIENCE

Software and AI Intern
Summer 2026
MS Solutions Group | SPARK payments platform
- Engineered real-time payment-transaction monitoring features using FastAPI, WebSockets, and Redis, including regional and top-store views, financial indicators, and fraud alerts.
- Implemented backend APIs and live data flows supporting user-facing transaction visualizations and fraud analytics.
- Integrated a Gemini API chatbot for conversational access to dashboard data and company information in a payments environment serving 300+ customers across 30+ countries.

Software Intern
2024-2025
Telnet
- Developed and deployed a Docker-based live metrics dashboard from supervisor specifications.

Web Development Intern
2023-2024
Shamash IT
- Developed user-facing web interfaces with HTML, CSS, and JavaScript.

ENTREPRENEURSHIP & INDEPENDENT PRODUCT DEVELOPMENT

Independent Software Developer & Entrepreneur
2026-Present
Business Portfolio: melek-business.vercel.app
- Established an independent software development initiative focused on helping local businesses modernize their digital presence, customer experience, and retention strategies.
- Designed and deployed branded, mobile-first interactive websites and QR-based digital menu concepts for restaurants and dessert businesses, including Slayta, Crumbz, and Overdozz.
- Developing a reusable, white-label customer loyalty platform combining digital membership, QR identification, customer accounts, personalized rewards, and business administration workflows.
- Implemented transactional email and account-verification functionality for the Slayta loyalty experience, connecting customer-facing interfaces with backend services.
- Conducted market research, developed product demonstrations, and presented solutions directly to prospective business clients, including an in-person Slayta demonstration that led to further discussions.
- Managed the end-to-end product development process, from requirements discovery and UI/UX design to implementation, cloud deployment, and iterative testing.

EDUCATION

ESPRIT - Engineering Degree in Computer Science, Artificial Intelligence
Expected 2028

Ranked **1st** in annual class ranking (2024-2025); 2nd (2023-2024). Final-year capstone (PFE) begins Fall 2027.

Baccalaureate in Mathematics, Lycee El Menzah 9
2023

SELECTED PROJECTS

Plant DNA Optimization and Expression Prediction
2025-2026
Deep Learning | Streamlit
- Built a Streamlit application for plant genetic-data analysis and expression prediction; selected for the Bal des Projets deep-learning showcase.
- Extended genomic analysis with host-aware codon optimization and grouped surrogate-model evaluation of simulated expression targets.

Reward Goblin - Reinforcement Learning Lab
2026 - ongoing
Python | PyTorch | Gymnasium | Stable-Baselines3 | Pymunk | FastAPI
- Built a physics playground that exposes reward hacking by comparing trained PPO policies with an intended-task reference controller.
- Evaluated reward versions across 3 training seeds, held-out starts and unseen layouts; separated task success from reward score and added heuristic exploit diagnostics.

Hydra - AI Coding Agent Orchestrator
2026 - ongoing
TypeScript | Electron | Node.js | MCP | Claude Code | Codex
- Built a desktop workspace with MCP tools to spawn, prompt, monitor and restart coding agents; added resumable sessions, headless runs and mobile remote control.
- Organized development around PLAN > IMPLEMENT > REVIEW > FIX > VERIFY > DEMO > APPROVAL, with controlled execution and human approval.

Telekinesis CV - Gesture-Controlled Computer Vision
2026 - ongoing
Python | OpenCV | MediaPipe | EdgeSAM | ONNX Runtime
- Built a CPU-only webcam application combining hand tracking and point-prompted segmentation to select and manipulate real-object appearances in video.
- Implemented pinch-to-grab, throwing, two-hand scaling and rotation, with background reconstruction and occlusion-aware compositing.

LogiXpress - Sustainable Delivery Platform
2024-2025
PHP | MySQL | MVC
- Developed an MVC delivery application with client, order, and database-backed workflows; won a Bal des Projets award.

WaveWorx - Operations Desktop Application
2024-2025
C++ | Qt | Oracle
- Implemented desktop modules for inventory and business records, including search, stock alerts, charts, and PDF reporting.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, PHP, SQL
Software / Backend: FastAPI, REST APIs, WebSockets, Redis, Django, Streamlit, HTML/CSS, Qt, MVC
Data: PostgreSQL, MySQL, MongoDB, Oracle, SQL / NoSQL
Infrastructure / Tools: Docker, Linux, Git; Node.js, Electron, MCP
AI / ML: Machine Learning, Deep Learning, Generative AI, Gemini API; PyTorch, PPO, Gymnasium, OpenCV, MediaPipe

CERTIFICATIONS & RECOGNITION

NVIDIA certificates in Machine Learning, Deep Learning, and Generative AI.
- **NVIDIA:** Fundamentals of Deep Learning
- **NVIDIA:** Generative AI with Diffusion Models
- **Amazon Web Services (Coursera):** Gen AI Dev- Agentic AI Solutions and Tool Integrations
- **Amazon Web Services (Coursera):** Gen AI Dev- Analyze Requirements & Design GenAI Solutions
- **Vanderbilt University (Coursera):** AI Agents and Agentic AI with Python & Generative AI
- Bal des Projets: winner for LogiXpress; selected for deep-learning, C++ application, and Linux game showcases.